

Generalization of the Alternating Product Limit

Based on the image “dailyintegral-limit-limit-1x1-2026-03-10.jpg”, the problem asks for the evaluation of the following infinite alternating product:

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{1}{k}\right)^{(-1)^k}$$

We can rigorously generalize this problem by replacing the constant 1 with an arbitrary complex variable z .

Theorem

For any $z \in \mathbb{C} \setminus \{-1, -2, -3, \dots\}$, the generalized infinite product converges to a closed-form expression utilizing the Gamma function:

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{z}{k}\right)^{(-1)^k} = \frac{\Gamma\left(\frac{z+1}{2}\right)}{\sqrt{\pi} \Gamma\left(\frac{z}{2} + 1\right)}$$

Rigorous Proof

Let $P_n(z)$ denote the n -th partial product of the sequence:

$$P_n(z) = \prod_{k=1}^n \left(1 + \frac{z}{k}\right)^{(-1)^k}$$

To evaluate the limit, we first examine the subsequence of even indices, $P_{2N}(z)$, by factoring the product into its numerator (even k terms, positive exponents) and denominator (odd k terms, negative exponents):

$$P_{2N}(z) = \frac{\prod_{j=1}^N \left(1 + \frac{z}{2j}\right)}{\prod_{j=1}^N \left(1 + \frac{z}{2j-1}\right)}$$

We can rewrite both the numerator and denominator using the fundamental property of the Gamma function, specifically that $\prod_{j=1}^N (j+\alpha) = \frac{\Gamma(N+1+\alpha)}{\Gamma(1+\alpha)}$.

1. Evaluating the Numerator

The product of the even terms can be expressed as:

$$\prod_{j=1}^N \left(1 + \frac{z}{2j}\right) = \frac{1}{N!} \prod_{j=1}^N \left(j + \frac{z}{2}\right) = \frac{\Gamma\left(N + 1 + \frac{z}{2}\right)}{\Gamma\left(1 + \frac{z}{2}\right) \Gamma(N + 1)}$$

2. Evaluating the Denominator

The product of the odd terms requires a slight algebraic manipulation first:

$$\prod_{j=1}^N \left(\frac{2j - 1 + z}{2j - 1}\right) = \frac{\prod_{j=1}^N \left(j - \frac{1}{2} + \frac{z}{2}\right)}{\prod_{j=1}^N \left(j - \frac{1}{2}\right)}$$

Applying the Gamma function identity to this quotient yields:

$$\frac{\Gamma\left(N + \frac{1}{2} + \frac{z}{2}\right) / \Gamma\left(\frac{1}{2} + \frac{z}{2}\right)}{\Gamma\left(N + \frac{1}{2}\right) / \Gamma\left(\frac{1}{2}\right)} = \frac{\Gamma\left(N + \frac{1}{2} + \frac{z}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{z+1}{2}\right) \Gamma\left(N + \frac{1}{2}\right)}$$

3. Ratio and Asymptotic Limit

Dividing the evaluated numerator by the denominator, we construct $P_{2N}(z)$:

$$P_{2N}(z) = \left[\frac{\Gamma\left(N + 1 + \frac{z}{2}\right)}{\Gamma(N + 1)} \frac{\Gamma\left(N + \frac{1}{2}\right)}{\Gamma\left(N + \frac{1}{2} + \frac{z}{2}\right)} \right] \frac{\Gamma\left(\frac{z+1}{2}\right)}{\Gamma\left(\frac{z}{2} + 1\right) \Gamma\left(\frac{1}{2}\right)}$$

To evaluate this as $N \rightarrow \infty$, we apply the standard asymptotic expansion for the ratio of Gamma functions, $\frac{\Gamma(N+a)}{\Gamma(N+b)} \sim N^{a-b}$:

- $\frac{\Gamma(N+1+\frac{z}{2})}{\Gamma(N+1)} \sim N^{z/2}$
- $\frac{\Gamma(N+\frac{1}{2})}{\Gamma(N+\frac{1}{2}+\frac{z}{2})} \sim N^{-z/2}$

The limit of the bracketed expression as $N \rightarrow \infty$ is exactly $N^{z/2} \cdot N^{-z/2} = 1$. Substituting the known value $\Gamma(1/2) = \sqrt{\pi}$ into the remaining constant terms provides the limit for the even subsequence:

$$\lim_{N \rightarrow \infty} P_{2N}(z) = \frac{\Gamma\left(\frac{z+1}{2}\right)}{\sqrt{\pi} \Gamma\left(\frac{z}{2} + 1\right)}$$

4. Convergence of the Odd Subsequence

To prove strict convergence for the entire sequence, we evaluate the odd indices $P_{2N+1}(z)$:

$$P_{2N+1}(z) = P_{2N}(z) \left(1 + \frac{z}{2N+1}\right)^{-1}$$

Because $\lim_{N \rightarrow \infty} \left(1 + \frac{z}{2N+1}\right)^{-1} = 1$, the odd subsequence converges to the exact same limit as the even subsequence. Thus, the general limit exists and is rigorously proven.

Application to the Original Problem

To solve the specific limit presented in the problem, we simply substitute $z = 1$ into our generalized derived formula:

$$L(1) = \frac{\Gamma\left(\frac{1+1}{2}\right)}{\sqrt{\pi} \Gamma\left(\frac{1}{2} + 1\right)} = \frac{\Gamma(1)}{\sqrt{\pi} \Gamma\left(\frac{3}{2}\right)}$$

Using the well-known specific values of the Gamma function, where $\Gamma(1) = 1$ and $\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}$:

$$L(1) = \frac{1}{\sqrt{\pi} \left(\frac{1}{2}\sqrt{\pi}\right)} = \frac{2}{\pi}$$

The exact value for the daily integral problem limit is $2/\pi$.